



AGENT SKILL ARCHITECTURE

Gemini Setup Skill

Fully Automated, Cross-Platform Onboarding & Diagnostics for Gemini
CLI & AI Studio Environments

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The Problem: Friction of API & CLI Setup

01

Manual Configuration

Developers spend high-effort cycles manually writing local environment variables, finding correct folders, and updating configuration JSON profiles.

02

Session Synchronization

Local environment variables set at the OS/registry level suffer from session latency, demanding terminal restarts before they are detectable by tools.

03

Security & Secret Leakage

Without automated protection mechanisms, developers frequently commit raw private API keys in active `.env` files onto public repositories.

The Solution: 6-Phase Orchestrated Skill

- ✓ **Phase 1: Diagnostics** - Automatically audits prerequisites (Node.js, NPM, CLI) and executes direct HTTPS API connection checks.
- ✓ **Phase 2: Installer** - Automatically downloads and installs missing dependencies globally without prompt blocking.
- ✓ **Phase 3: Copilot Guidance** - Uses Chrome CDP redirection to safely guide the user to their private API key console, dodging bot-protection.
- ✓ **Phase 4: Persistence** - Safely writes keys to multiple fail-safes (.env, user profile, registry) with immediate shell process injection.
- ✓ **Phase 5: Verification** - Instantly tests the environment using live CLI generation prompts to verify complete end-to-end integration.
- ✓ **Phase 6: Self-Healing** - Intercepts billing, 429 exhaustion, and region errors, providing targeted instructions.

The Synergy: Google Cloud, Antigravity & Gemini



Google Cloud

- ✦ **Vertex AI & AI Studio Platform:** Hosted the foundation model endpoints (Gemini 2.5/3.5 Flash) ensuring secure, ultra-low latency response times.
- ✦ **Secure IAM Integration:** Managed cloud credentials, security keys, user scopes, and Application Default Credentials (ADC).
- ✦ **Prepayment Quotas:** Offered robust API balance structures with automatic limit barriers for developers.

INFRASTRUCTURE LAYER



Antigravity (Agent)

- ✦ **Automated Skill Compiler:** Architected the local cross-platform `gemini_setup` automated skill.
- ✦ **Zero-Dependency Diagnostics:** Created `check_env.js` for instant shell environmental and connectivity testing.
- ✦ **Chrome CDP Integration:** Commanded local browser redirects to safely acquire private keys under security challenges.
- ✦ **Immediate Session Injector:** Programmed direct shell registry/rc injections to bypass terminal restart lags.

INTELLIGENT ORCHESTRATOR



Gemini LLM

- ✦ **Terminal Assistant CLI Engine:** Powered `@google/gemini-cli` to parse and execute live terminal tasks.
- ✦ **Context-Aware Dialogue:** Conducted real-time shell-integrated developer chats with extreme comprehension.
- ✦ **Self-Healing Advisory:** Analyzed logical errors and dynamically recommended diagnostic solutions (e.g. Quota/429 issues).

INTELLIGENCE LAYER

Multi-Level Persistence System

1. Local Fallback (Project .env)

Configured locally under the active project directory for localized development.

2. Global Backup (Home .env)

Stored at `~/.env` to act as a fallback option across directory shifts.

3. Tool Configuration (settings.json)

Directly integrates API auth types into `~/.gemini/settings.json`.



Persistent Registry & Session Injection

Ensures full cross-platform persistence and activates variables instantly in the active terminal process without requiring shell restarts.

```
# Windows (PowerShell) Registry & Current Session
[System.Environment]::SetEnvironmentVariable('GEMINI_API_KEY', '<key>', 'User')
$env:GEMINI_API_KEY = '<key>'

# Unix (macOS / Linux) Shell rc & Session
echo 'export GEMINI_API_KEY="<key>"' >> ~/.zshrc
export GEMINI_API_KEY="<key>"
```

Robust Zero-Dependency Diagnostics

The diagnostics script `check_env.js` compiles Node's built-in `fs`, `path`, and `os` libraries. It automatically loads credentials without third-party libraries (no `dotenv` required), eliminating setup dependencies.

```
// Built-in .env Loader inside check_env.js
function loadEnvFile(filePath) {
  if (fs.existsSync(filePath)) {
    const content = fs.readFileSync(filePath, 'utf-8');
    content.split(/\r?\n/).forEach(line => {
      // Parses keys and automatically maps to process.env
    });
  }
}
```

Visual Diagnostic Report Output

```
=== Checking Prerequisites & Dependencies ===
[OK] Node.js (v22.18.0)
[OK] NPM (11.5.2)
[OK] Gemini CLI (@google/gemini-cli) (0.49.0)

=== Checking Environment Variables ===
[OK] GEMINI_API_KEY (AI Studio) (Set)

=== Testing API Connectivity ===
[OK] Google AI Studio Access (Gemini 2.5 Flash)
```

Billing & Quota Self-Healing

Quota Depleted Error (RESOURCE_EXHAUSTED)

Matches the exact error string where developers hit a 429 because their Google AI Studio prepayment credit balance is empty.

```
"error": {  
  "code": 429,  
  "message": "Your prepayment credits are depleted..."  
}
```

Step-by-Step Resolution Path

- ✓ **Open Studio Settings** - Guide the developer straight to Google AI Studio billing consoles.
- ✓ **Deposit Balance** - Recommends topping up a minor prepayment balance (e.g. \$5) to release rate limits.
- ✓ **Revert to Free Tier** - Provides options to map the API key to a Free Tier project to run on standard quota.

Open Agent Skills Ecosystem Integration

Fully packaged and prepared to be installed across diverse developer machines and AI agents instantly using the standardized skills CLI.

```
npx skills add hextrump/geminisetupapi@gemini_setup -g -y
```

✓ 100% compliant with Open Agent Skills
Spec

✓ Installs globally for all assistant
profiles

✓ Complete and ready in under 10
pages